

Fig. 5: Typical setup of an IR water faucet

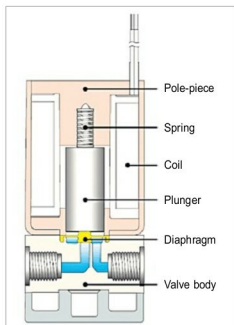


Fig. 6: Structure of a solenoid valve

bined, constitute an IR water faucet. Here is an overview breakdown of the components:

1. IR water faucets employ presence sensors (not motion sensors) to sense the presence of an object in front of the faucet. These employ active IR technology that senses pres-

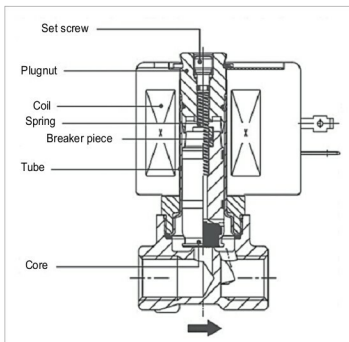


Fig. 7: Parts of a proportional solenoid valve

ence, and not movement, of objects. Active IR technology, like the name implies, actively emits IR light and actively waits for this light to come back to it.

2. Based on sensor feedback, the electronic control circuitry commands the solenoid valve to start/stop the flow of water.

3. Faucets powered by AC transformers utilise standard solenoid valves. Such solenoid valves require constant energy to hold the plunger in place, and when e-energised, the plunger naturally returns to its normal closed position with the help of a biasing spring.

On the other hand, faucets powered by batteries employ another type of solenoid valve called pulse or latching solenoid valve. These solenoid valves operate at low voltage, usually 6V DC, with some solenoids operating at 9V DC.

The reason these solenoids are called latching type is that, as the solenoid is initially energised to start the water flow, the plunger is driven into the range of a permanent magnet, which, in turn, holds the plunger in open position. This initial energising of the solenoid is called pulsing and takes place within a

fraction of a second. In order to return the plunger to its original closed position, the solenoid is once again pulsed, but this time by reversing polarity. This complex operation, however, can conserve battery power.

4. The solenoid valve physically controls the start and stop of the water flow through the faucet spout. The solenoid valve is, in fact, a perfect blend of two hybrid

technologies—electromagnetism (solenoid) and fluid dynamics (diaphragm valve).

A solenoid is an electrical component that transforms electrical energy into mechanical energy—motion. When energised, it creates a magnetic field that exerts a linear force on an object called plunger. The basic premise behind the diaphragm valve is to control the flow of a large volume of water with a smaller, more manageable water volume. In order to open the diaphragm valve, there is need to release the pressure that is pinning the diaphragm against the valve seat—this is the job of the solenoid valve.

The plunger, which is driven by the solenoid, is yet another miniature water valve whose sole mission is to open and release the pressure that is pinning the diaphragm against the valve seat. It closes and allows the pressure supplied through a very small diameter hole to build up behind the diaphragm, pinning the diaphragm back against the valve seat.

5. Proportional solenoid valves are a newcomer in the solenoid valve family. Most solenoid valves work on an on/off basis, and are either